

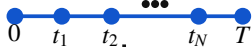
PINN representation of the controlled evolution

(a1) time labels and T

normalized time $s_i \in [0, 1]$



physical time $t_i = s_i T$

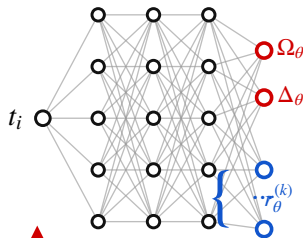


τ (trainable)

$$T = e^\tau$$

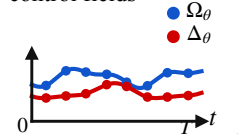
(b1) PINN generator

f_θ

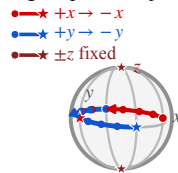


(c1) outputs

control fields

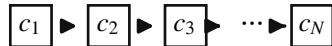


Z-gate probe trajectories

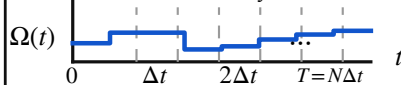


Discrete pulse optimization

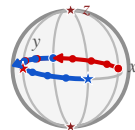
(a2) pulse amplitudes



T fixed or externally scanned



(b2) propagator



$$\mathbf{r}(T) = \Phi(T)\mathbf{r}(0)$$

$$\Phi(T) = P \exp \left[\int_0^T F(t) dt \right]$$